

Statics Problems Handout¹

1. Compute the total resultant force, $\mathbf{R}$, and resultant moment, $\mathbf{M}_R^{(0)}$ (about the origin), for the two-dimensional object shown in Fig. 1.

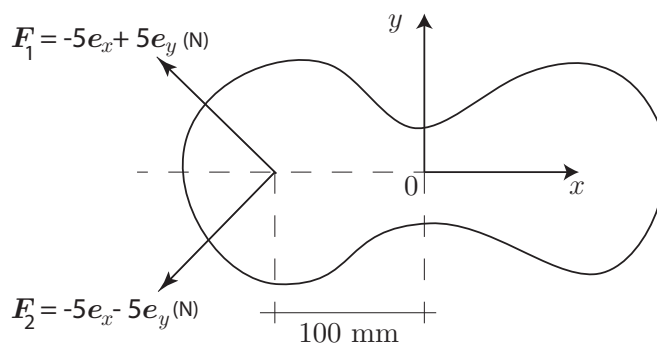


Figure 1: Problem 1.

2. In Fig. 2 a collar that can slide on a vertical rod is subjected to three applied forces. Find the angle α such that the resultant force being applied to the collar is horizontal; i.e. determine the condition such that $\mathbf{R} \cdot \mathbf{e}_y = 0$. If you do not see how to solve this problem exactly, it is perfectly ok to determine your angle by trial and error, by making a plot, or solving using a numerical algorithm, et cetera.
3. The hook in Fig. 3 is subjected to three forces as show, where $\alpha = \pi/6$ and $\beta = \pi/4$. Determine the magnitude and direction of the resultant of the three applied forces.
4. A large cylinder with weight $4W$ rests upon two smaller cylinders with weight W as shown in Fig. 4. Assume that all the surfaces are smooth and frictionless so that the contact forces are all orthogonal to the contact surfaces. Further assume that the cylinders are homogeneous so that their weight can be assumed to act at their centers

¹In preparing solutions, you are free to use any available tools, e.g. Matlab, Python, Mathematica, calculator, etc. It is my philosophy that you should *always* use the best and most efficient tools available to solve a given problem, unless I specify otherwise. Further, graphs should be properly plotted using Matlab, Python, Mathematica, etc. with properly labeled axes. Sketches should always be neat and professional. Be very cautious about using large language models. The point of the homework is for you to develop your skills and understanding. That takes diligent practice and sometimes painful struggles. **With the exception of graphs, homework solutions must be hand-written.**

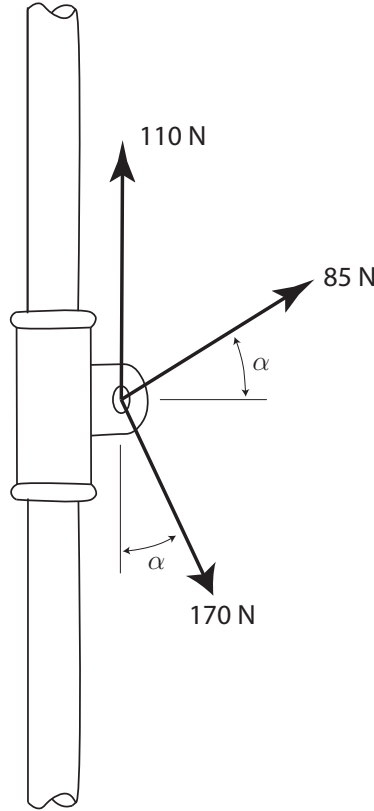


Figure 2: Problem 2.

in the downward direction. Find the contact forces at all points of contact. [Hint: The contact point between the large cylinder and the small cylinder occurs on the line that joins the centers of the cylinders.]

5. A massless rod of length l is inserted in a slot and subjected to a vertical force of magnitude F ; see Fig. 5. Assuming that the rod is in a state of static equilibrium, find the necessary relation between a and l for this to hold true under the assumption that all surfaces are smooth (frictionless).
6. A massless plate of dimensions $b \times 17.5a$ is suspended by three vertical wires and is in static equilibrium; see Fig. 6. Find the location where the force F needs to be applied in order for the wires to all carry the same load.
7. Consider a unit cube subjected to a set of loads that can be characterized in a point A , $\mathbf{r}_{0A} = \mathbf{e}_x + \mathbf{e}_y + \mathbf{e}_z(m)$, by $\{\mathbf{R}, \mathbf{M}_R^{(A)}\} = \{2\mathbf{e}_x + 3\mathbf{e}_z(N), 5\mathbf{e}_y + 2\mathbf{e}_z(N \cdot m)\}$. The cube is supported by a set of hinges as shown in Fig. 7. Find the moment that the loads create to rotate the cube along the axis connecting the hinges.
8. The body shown in Fig. 8 is subjected to two force systems. The first is characterized

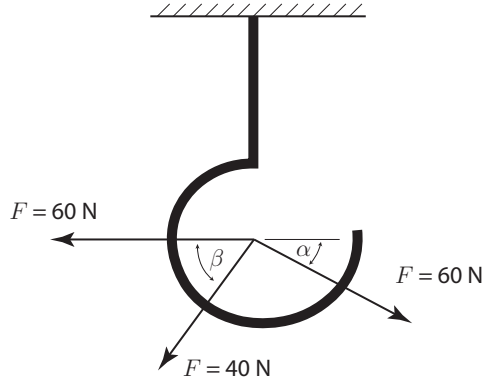


Figure 3: Problem 3.

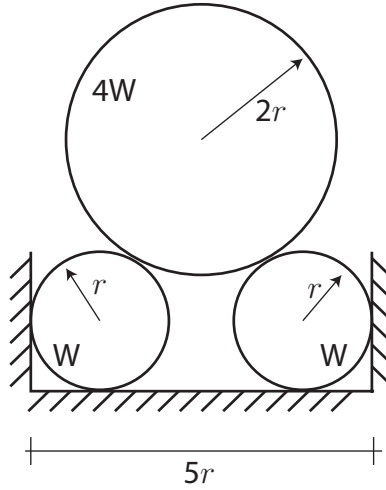


Figure 4: Problem 4.

by the resultants $\{\mathbf{F}_A, \mathbf{M}_R^{(A)}\}$ and the second by the resultants $\{\mathbf{F}_B, \mathbf{M}_R^{(B)}\}$. If the body is in static equilibrium, find the necessary relations between $\mathbf{F}_A$, $\mathbf{F}_B$, $\mathbf{M}_R^{(A)}$, and $\mathbf{M}_R^{(B)}$. Assume that point A is located at $\mathbf{r}_{0A}$ and point B is located at $\mathbf{r}_{0B}$.

9. For the beam shown in Fig. 9, find the support reactions.
10. If the magnitude of the support force for the S-bracket in Fig. 10 is 1000 N, what is the vectorial expression for the force at A? Assume that the S-bracket is in static equilibrium.
11. The plate in Fig. 11 is subjected to three vertical forces of magnitude $F_1 = 25$ N, $F_2 = 50$ N, $F_3 = 75$ N. Find an equivalent single force system to replace these three forces. Make sure to specify the location where the equivalent force acts.

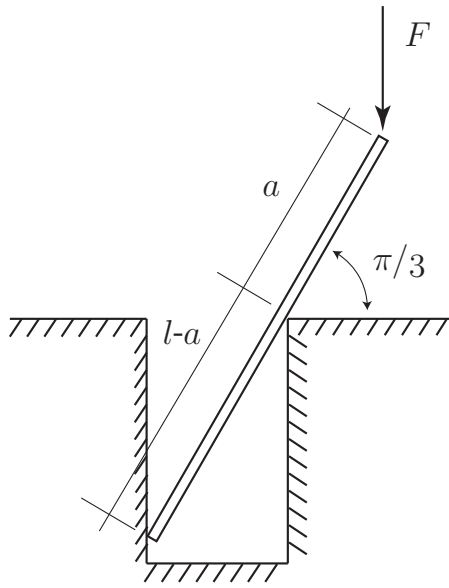


Figure 5: Problem 5.

12. The unit cube in Fig. 12 is subjected to a pair of 1 kN forces at A and B . Can this force system be replaced by a single point force? If so, at what point does it act? If not, why is that?
13. The beam in Fig. 13 is subjected to a distributed load $q(x) = Ax^{1/2}$ (Force/Length). Find the equivalent point force system to this distributed load, its magnitude and the location where it acts.
14. A plate as shown in Fig. 14 is subjected to a distributed normal pressure load $p(x, y) = Axy^2$. Find the equivalent point force system to this distributed load, its magnitude and the location where it acts.
15. For the structure in Fig. 15 determine the support reactions at A and the forces in bars 1, 2, and 3.
16. For the truss in Fig. 16 determine the forces in bars BD and DE.
17. A 300 N force acts on a 1000 N crate that rests on an inclined plane; see Fig. 17. The coefficient of static friction between the crate and the plane is $\mu_0 = 0.25$ and the coefficient of kinetic friction is $\mu = 0.2$. Determine if the crate is in a state of static equilibrium and find the resultant force that the plane applies to the crate (including its location).

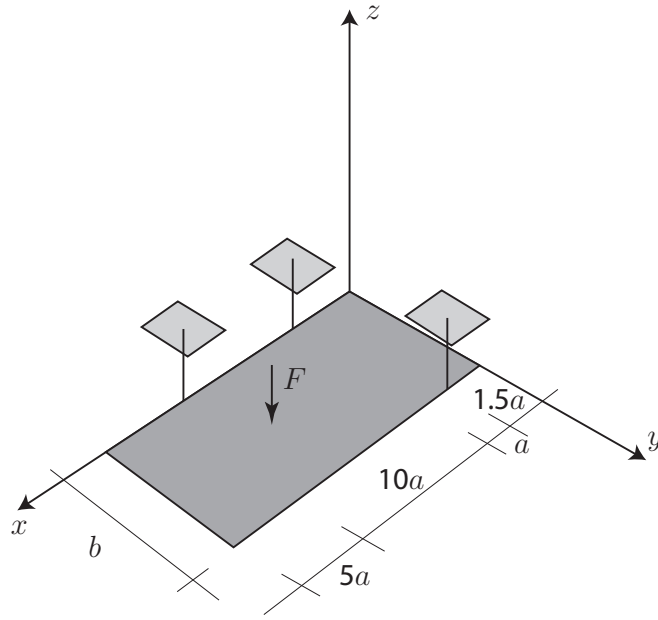


Figure 6: Problem 6.

18. The cabinet in Fig. 18 weighs 500 N with center of gravity as show. The cabinet is supported on two casters. Caster A is unlocked and free to roll. Caster B is locked and has a static coefficient of friction with the floor of $\mu_0 = 0.3$. What force F is required to slide the cabinet? Does the force depend on the height h where it is applied? If h is too large the cabinet can tip over. Determine this height.

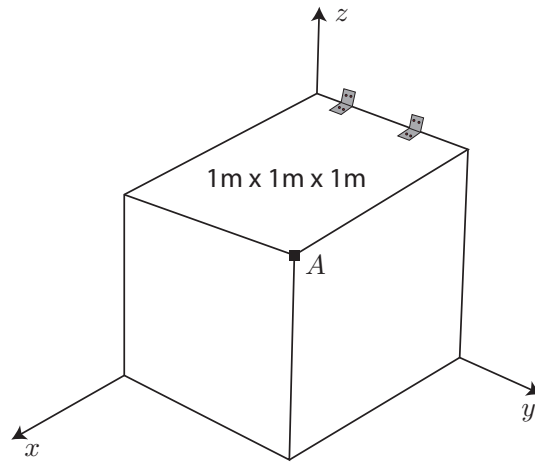


Figure 7: Problem 7.

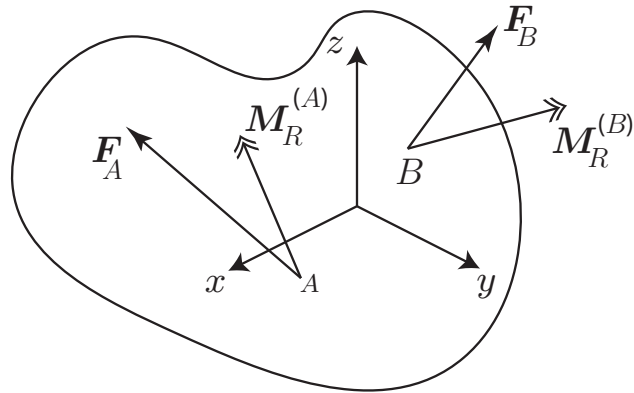


Figure 8: Problem 8.

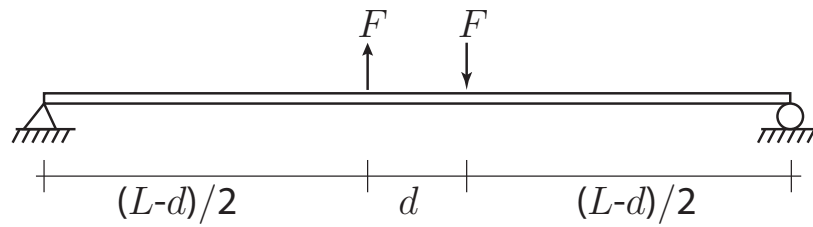


Figure 9: Problem 9.

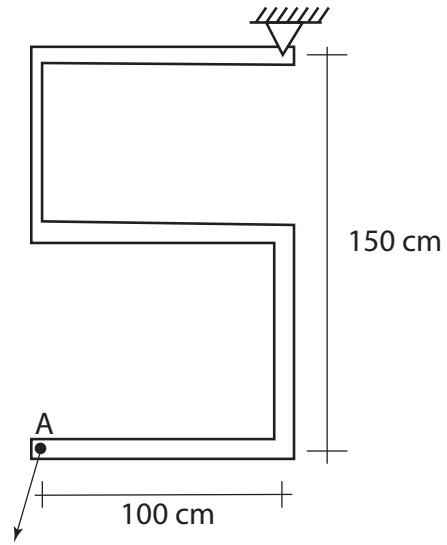


Figure 10: Problem 10.

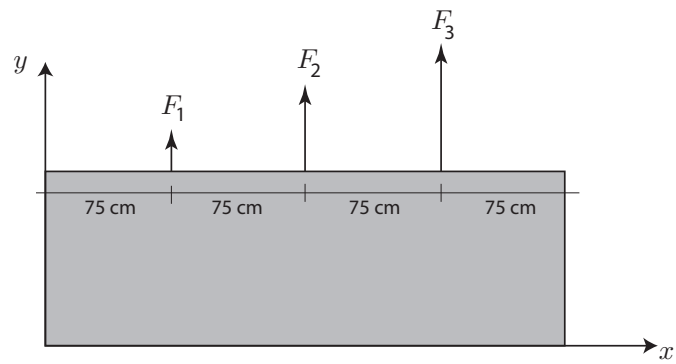


Figure 11: Problem 11.

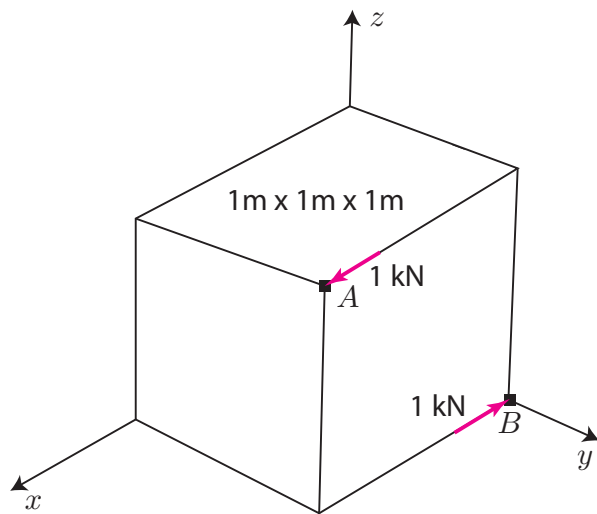


Figure 12: Problem 12.

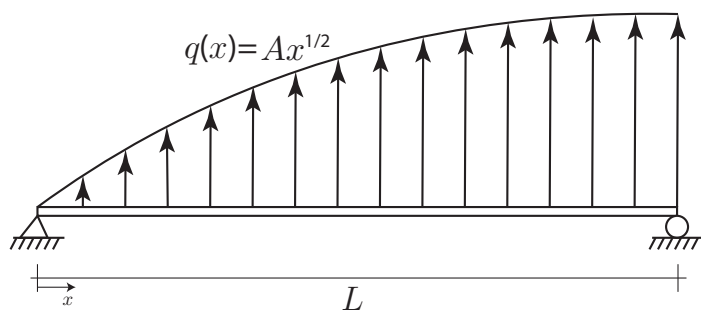


Figure 13: Problem 13.

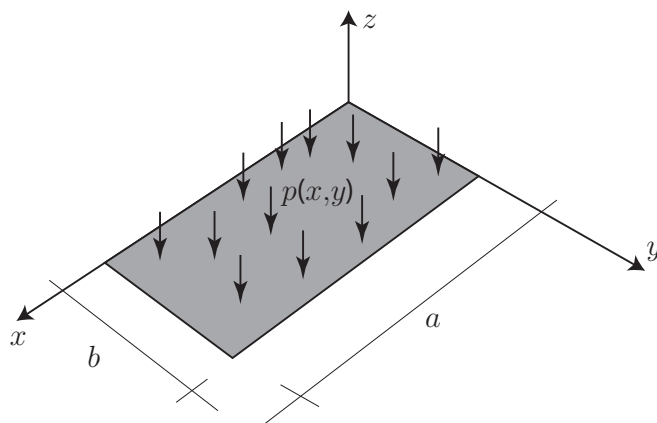


Figure 14: Problem 14.

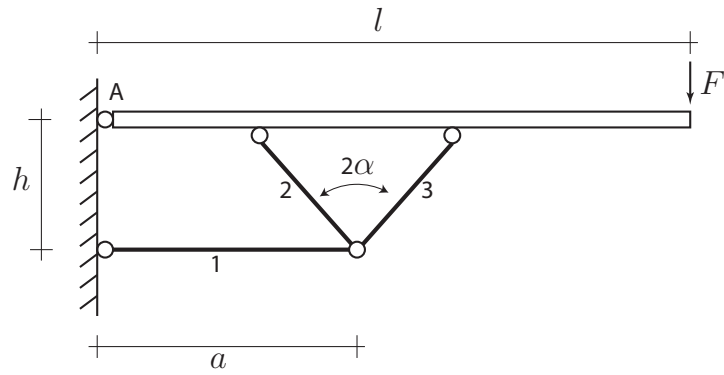


Figure 15: Problem 15.

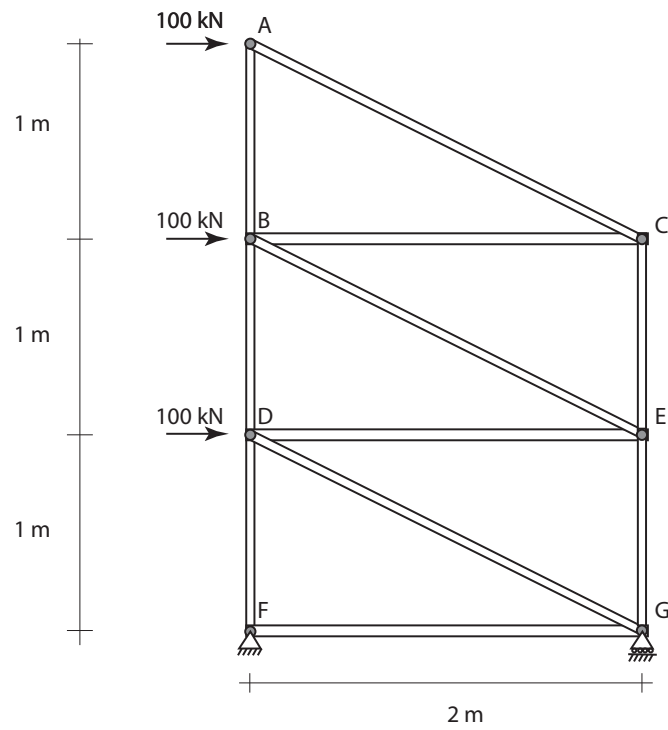


Figure 16: Problem 16.

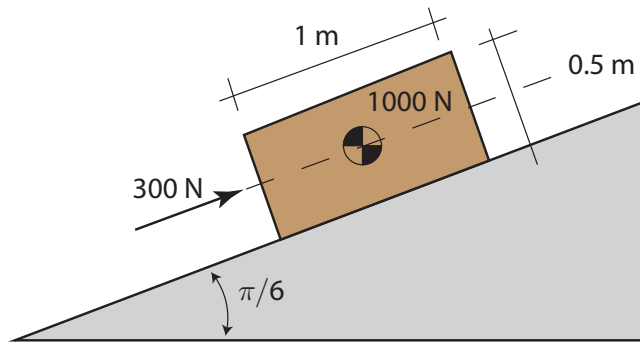


Figure 17: Problem 17.

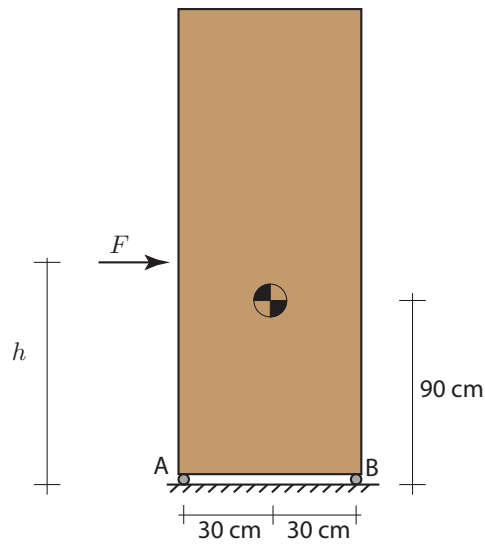


Figure 18: Problem 18.